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A FIRST COURSE ON DIFFERENTIAL EQUATIONS WITH MODELING APPLICATION BY DENNIS G. ZILL (10e)

LIST OF RESULTS

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1 Introduction to Differential Equations

Theorem 1.1. *(existence of a unique solution). Let R be a rectangular region in the xy-plane defined by $a \leq x \leq b$, $c \leq y \leq d$ that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f/\partial y$ are continuous on R, then there exists some interval $I_0 : (x_0 - h, x_0 + h)$, $h > 0$, contained in $[a, b]$, and a unique function $y(x)$ defined on I_0, that is a solution of the initial-value problem.*

2 First-Order Differential Equations

Theorem 2.4.1 (*Criterion for an Exact Differential*) Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous partial derivatives on a rectangle R in the xy-plane. Then a necessary and sufficient condition that $M(x, y)dx + N(x, y)dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \tag{1}$$

3 Higher-Order Differential Equations

Theorem 3.1. *(existence of a unique solution). Let $a_n(x), a_{n-1}(x), \ldots, a_0(x)$ and $g(x)$ be continuous on an interval I and let $a_n(x) \neq 0$ for every x in I. If $x = x_0$ is any point in I, then a solution $y(x)$ of the initial-value problem exists on I and is unique.*

Theorem 3.2. *(superposition principle—homogeneous equations). Let $y_1(x), y_2, \ldots, y_k(x)$ be solutions of a homogeneous n-th order linear differential equation. Then the linear combination*

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x), \tag{2}$$

where c_i, $i = 1, 2, \ldots, k$ are arbitrary constants, is also a solution of the equation.

Theorem 3.3. *(criterion for linearly independent solutions). Let $y_1(x), y_2(x), \ldots, y_n(x)$ be n solutions of a homogeneous n-th order linear differential equation on an interval I. Then the set of solutions is linearly independent on I iff the Wronskian $W(y_1, y_2, \ldots, y_n) \neq 0$ for every x in I, where*

$$W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}. \tag{3}$$

Theorem 3.4. *(existence of a fundamental set). There exists a fundamental set of solutions for the homogeneous n-th order linear differential equation on an interval I.*

Theorem 3.5. *(general solution—homogeneous equations). Let $y_1(x), y_2(x), \ldots, y_n(x)$ be a fundamental set of solutions of a homogeneous n-th order linear differential equation on an interval I. Then the* general solution *of the equation on the interval is given by*

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x), \tag{4}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Theorem 3.6. *(general solution–nonhomogeneous equations). Let $y_p(x)$ be any particular solution of a nonhomogeneous n-th order linear differential equation on an interval I and let $y_1, y_2, \ldots, y_n$ be a fundamental set of solutions of the associated homogeneous equation on the same interval. Then the* general solution *of the nonhomogeneous equation on the interval is given by*

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x), \tag{5}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Theorem 3.7. *(superposition principle—nonhomogeneous equations). Let $y_{p1}, y_{p2}, \ldots, y_{pk}$ be k particular solutions of a nonhomogeneous n-th order linear differential equation on an interval I corresponding, in turn, to k distinct functions $g_1(x), g_2(x), \ldots, g_k(x)$. That is, suppose y_{pi} denotes a particular solution of the corresponding differential equation*

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g_i(x), \quad i = 1, 2, \ldots, k. \tag{6}$$

then

$$y_p(x) = y_{p1}(x) + y_{p2}(x) + \cdots + y_{pk}(x) \tag{7}$$

is a particular solution of

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g_1(x) + g_2(x) + \cdots + g_k(x). \tag{8}$$

Theorem 3.8. *(reduction of order). Let $y_1(x)$ be a solution of the homogeneous n-th order linear differential equation on an interval I and that $y_1(x) \neq 0$ for all x in I. Then*

$$y_2(x) = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx \tag{9}$$

is a second solution of the equation on the interval.

4 The Laplace Transform

Definition 4.1. (Laplace transform). Let f be a function defined on $[0, \infty)$. Then the function F defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)dt \tag{10}$$

is said to be the *Laplace transform* of f. The domain of F is the set of all s for which the improper integral converges.

Theorem 4.1. *(transforms of some basic functions).*

(a) $\mathcal{L}\{1\} = \dfrac{1}{s}$

(b) $\mathcal{L}\{t^n\} = \dfrac{n!}{s^{n+1}}$ *for* $n = 0, 1, 2, \ldots$

(c) $\mathcal{L}\{e^{at}\} = \dfrac{1}{s-a}$

(d) $\mathcal{L}\{\sin(\omega t)\} = \dfrac{\omega}{s^2+\omega^2}$

(e) $\mathcal{L}\{\cos(\omega t)\} = \dfrac{s}{s^2+\omega^2}$

(f) $\mathcal{L}\{\sinh(\omega t)\} = \dfrac{\omega}{s^2-\omega^2}$

(g) $\mathcal{L}\{\cosh(\omega t)\} = \dfrac{s}{s^2-\omega^2}$

Definition 4.2. (exponential order). A function f defined on $[0, \infty)$ is said to be of *exponential order* as $t \to \infty$ if there exist constants c, $M > 0$, and $T > 0$ such that $|f(t)| \leq Me^{ct}$ for all $t > T$. If f is of exponential order, then the Laplace transform $\mathcal{L}\{f(t)\} \leq Me^{ct}$ for all $t > T$.

Theorem 4.2. *(sufficient conditions for existence). If f is piecewise continuous on $[0, \infty)$ and of exponential order as $t \to \infty$, then the Laplace transform $\mathcal{L}\{f(t)\}$ exists for all $s > c$, where c is a constant such that $|f(t)| \leq Me^{ct}$ for all $t > T$.*

Theorem 4.3. *(behavior as $F(s)$ as $s \to \infty$). If f is piecewise continuous on $[0, \infty)$ and of exponential order as $t \to \infty$ and $\mathcal{L}\{f(t)\} = F(s)$, then $\lim_{s\to\infty} F(s) = 0$.*

Theorem 4.4. *(some inverse transforms).*

(a) $\mathcal{L}^{-1}\left\{\dfrac{1}{s}\right\} = 1$

(b) $\mathcal{L}^{-1}\left\{\dfrac{n!}{s^{n+1}}\right\} = t^n$ *for* $n = 0, 1, 2, \ldots$

(c) $\mathcal{L}^{-1}\left\{\dfrac{1}{s-a}\right\} = e^{at}$

(d) $\mathcal{L}^{-1}\left\{\dfrac{\omega}{s^2+\omega^2}\right\} = \sin(\omega t)$

(e) $\mathcal{L}^{-1}\left\{\dfrac{s}{s^2+\omega^2}\right\} = \cos(\omega t)$

(f) $\mathcal{L}^{-1}\left\{\dfrac{\omega}{s^2-\omega^2}\right\} = \sinh(\omega t)$

(g) $\mathcal{L}^{-1}\left\{\dfrac{s}{s^2-\omega^2}\right\} = \cosh(\omega t)$

Theorem 4.5. *(transform of a derivative). Suppose f, f', ..., f^{n-1} are continuous on $[0,\infty)$, and f, f', ..., f^{n-1} are of exponential order as $t \to \infty$. Then for $s > c$,*

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - f^{(n-1)}(0) \tag{11}$$

where $F(s) = \mathcal{L}\{f(t)\}$.

Theorem 4.6. *(first translation theorem). If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > c$ and a is any real rumber, then*

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a) \tag{12}$$

for $s > c + a$.

Theorem 4.7. *(second translation theorem). If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > c \geq 0$, and a is a positive constant, then*

$$\mathcal{L}\{f(t-a)\,\mathcal{U}(t-a)\} = e^{-as}F(s) \tag{13}$$

for $s > c$.

Theorem 4.8. *(derivatives of transforms). If the function f is piecewise continuous on $[0,\infty)$, of exponential order and $\mathcal{L}\{f(t)\} = F(s)$, then for $n = 1, 2, 3, \ldots$ and $s > c$,*

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}F(s). \tag{14}$$

Theorem 4.9. *(convolution theorem). If the functions f and g are piecewise continuous on $[0,\infty)$, of exponential order and $\mathcal{L}\{f(t)\} = F(s)$, $\mathcal{L}\{g(t)\} = G(s)$, then for $s > c$,*

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\} = F(s)G(s). \tag{15}$$

Theorem 4.10. *(transform of a periodic function). If f is piecewise continuous $[0,\infty)$, of exponential order, and periodic with period T, then*

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}}\int_0^T e^{-st}f(t)dt. \tag{16}$$

Theorem 4.11. *(transform of the dirac delta function). For $t_0 > 0$,*

$$\mathcal{L}\{\delta(t-t_0)\} = e^{-st_0}. \tag{17}$$

5 Systems of Linear Differential Equations

Definition 5.1. (solution vector). A *solution vector* on an interval I is any column matrix

$$\mathbf{X} = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} \tag{18}$$

whose entries are differentiable functions satisfying the system $\mathbf{X}' = \mathbf{AX} + \mathbf{F}$ on the interval.

Theorem 5.1. *(existence of a unique solution). Let the entries of the matrices $\mathbf{A}(t)$ and $\mathbf{F}(t)$ be functions continuous on a common interval I that contains the point t_0. Then there exists a unique solution of the initial-value problem $\mathbf{X}' = \mathbf{A}(t)\mathbf{X} + \mathbf{F}(t)$, $\mathbf{X}(t_0) = \mathbf{X_0}$ on the interval.*

Theorem 5.2. *(superposition principle). Let $\mathbf{X}_1$, $\mathbf{X}_2$, ..., $\mathbf{X}_k$ be a set of solution vectors of the nonhomogeneous system $\mathbf{X}' = \mathbf{AX} + \mathbf{F}$ on an interval I. Then the linear combination*

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_k\mathbf{X}_k, \tag{19}$$

where c_i, $i = 1, 2, \ldots, k$ are arbitrary constants, is also a solution of the system on the interval.

Theorem 5.3. *(criterion for linearly independent solutions). Let* $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ *be* n *solution vectors of a homogeneous system on an interval* I*. Then the set of solution vectors is linearly independent on* I *iff the Wronskian*

$$W(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n) = \begin{vmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{vmatrix} \neq 0 \tag{20}$$

for every t *in the interval.*

Theorem 5.4. *(existence of a fundamental set). There exists a fundamental set of solutions for the homogeneous system* $\mathbf{X}' = \mathbf{AX}$ *on an interval* I*.*

Theorem 5.5. *(general solution–homogeneous systems). Let* $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ *be a fundamental set of solutions of a homogeneous system on an interval* I*. Then the* general solution *of the system on the interval is given by*

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_n\mathbf{X}_n, \tag{21}$$

where the c_i*,* $i = 1, 2, \ldots, n$ *are arbitrary constants.*

Theorem 5.6. *(general solution–nonhomogeneous systems). Let* $\mathbf{X}_p$ *be any particular solution of a nonhomogeneous system on an interval* I *and let* $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ *be a fundamental set of solutions of the associated homogeneous system on the same interval. Then the* general solution *of the nonhomogeneous system on the interval is given by*

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_n\mathbf{X}_n + \mathbf{X}_p, \tag{22}$$

where the c_i*,* $i = 1, 2, \ldots, n$ *are arbitrary constants.*

Theorem 5.7. *(distinct real eigenvalues: general solution—homogeneous systems). Let* $\lambda_1, \lambda_2, \ldots, \lambda_n$ *be* n *distinct real eigenvalues of the coefficient matrix* $\mathbf{A}$ *of the homogeneous system* $\mathbf{X}' = \mathbf{AX}$ *and let* $\mathbf{K}_1, \mathbf{K}_2, \ldots, \mathbf{K}_n$ *be the corresponding eigenvectors. Then the* ***general solution*** *of the system on the interval* $(-\infty, \infty)$ *is given by*

$$\mathbf{X} = c_1\mathbf{K}_1e^{\lambda_1 t} + c_2\mathbf{K}_2e^{\lambda_2 t} + \cdots + c_n\mathbf{K}_ne^{\lambda_n t}. \tag{23}$$

Theorem 5.8. *(solutions corresponding to a repeated eigenvalue). Let* λ_1 *be a real eigenvalue of multiplicity* k *of the coefficient matrix* $\mathbf{A}$ *of the homogeneous system* $\mathbf{X}' = \mathbf{AX}$ *and let* $\mathbf{K}_1$ *be an eigenvector corresponding to* λ_1*. Then the* k *linearly independent solutions of the system corresponding to the eigenvalue* λ_1 *are given by*

$$\mathbf{X}_1 = \mathbf{K}_1e^{\lambda_1 t} \tag{24}$$

$$\mathbf{X}_2 = t\mathbf{K}_1e^{\lambda_1 t} \tag{25}$$

$$\vdots \tag{26}$$

$$\mathbf{X}_k = t^{k-1}\mathbf{K}_1e^{\lambda_1 t} \tag{27}$$

Theorem 5.9. *(real solutions corresponding to complex eigenvalues). Let* $\lambda_1 = \alpha + i\beta$ *be a complex eigenvalue of the coefficient matrix* $\mathbf{A}$ *of the homogeneous system* $\mathbf{X}' = \mathbf{AX}$ *and let* $\mathbf{K}_1$ *be a corresponding eigenvector. Then the two linearly independent solutions of the system on the interval* $(-\infty, \infty)$ *are given by*

$$\mathbf{X}_1 = Re\left\{\mathbf{K}_1e^{\lambda_1 t}\right\} = e^{\alpha t}\left[Re\{\mathbf{K}_1\}\cos(\beta t) - Im\{\mathbf{K}_1\}\sin(\beta t)\right] \tag{28}$$

$$\mathbf{X}_2 = Im\left\{\mathbf{K}_1e^{\lambda_1 t}\right\} = e^{\alpha t}\left[Re\{\mathbf{K}_1\}\sin(\beta t) + Im\{\mathbf{K}_1\}\cos(\beta t)\right] \tag{29}$$

6 Dynamics of First-Order Difference Equations

The following is an introduction to difference equations, based on *An Introduction to Difference Equations* (3e), by S. Elaydi.

Definition 6.1. A linear *homogeneous* first-order difference equation is given by

$$x(n+1) = a(n)x(n), \quad x(n_0) = x_0, \quad n \geq n_0 \geq 0, \tag{30}$$

and the associated *nonhomogeneous* equation is

$$y(n+1) = a(n)y(n) + g(n), \quad y(n_0) = y_0, \quad n \geq n_0 \geq 0, \tag{31}$$

where in both cases $a(n) \neq 0$, and $a(n)$ and $g(n)$ are real-valued functions defined for $n \geq n_0$.

Result 6.1. *The solution to* (30) *can be obtained by iteration and mathematical induction and is given by*

$$x(n) = \left[\prod_{i=n_0}^{n-1} a(i)\right] x_0 \tag{32}$$

For the nonhomogeneous case, the unique solution is

$$y(n) = \left[\prod_{i=n_0}^{n-1} a(i)\right] y_0 + \sum_{r=n_0}^{n-1} \left[\prod_{i=r+1}^{n-1} a(i)\right] g(r). \tag{33}$$

Definition 6.2. (equilibrium point). A point x^* in the domain of f is said to be an *equilibrium point* of a difference equation

$$x(n+1) = f(x(n)) \tag{34}$$

if it is a fixed point of f, i.e., $f(x^*) = x^*$.

Definition 6.3. (eventually equilibrium point). Let x be a point in the domain of f. If there exists a positive integer r and an equilibrium point x^* of (34) such that $f^r(x) = x^*$, $f^{r-1} \neq x^*$, then x is an *eventually equilibrium (fixed) point.*

Definition 6.4. (a) The equilibrium point x^* of (34) is *stable* if given $\epsilon > 0$ there exists $\delta > 0$ such that $|x_0 - x^*| < \delta$ implies $|f^n(x_0) - x^*| < \epsilon$ for all $n > 0$. If x^* is not stable, then it is called *unstable.*

(b) The point x^* is said to be *attracting* if there exists $\eta > 0$ such that

$$|x(0) - x^*| < \eta \implies \lim_{n\to\infty} x(n) = x^*.$$

If $\eta = \infty$, x^* is called a *global attractor* or *globally attracting.*

(c) The point x^* is an *asymptotically stable equilibrium point* if it is stable and attracting. If $\eta = \infty$, x^* is said to be *globally asymptotically stable.*